Sarah Henia



Software Engineer

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Portfolio | LinkedIn | GitHub

SUMMARY

Software engineer who started in full-stack JavaScript (MERN) and now builds production systems in C#/.NET, Angular and Python: an enterprise real-time presence library, data pipelines, and AI agents deployed on AWS and Azure. Cares about code that other teams can install, test, trust and maintain.

EXPERIENCE

The SamurAI — *Remote, USA*
Software Architect & AI Solution Integrator — *Feb 2026 - Present*

- Designed and built **Dawn**, an autonomous **AI agent** for project management, live in production from March 2026: turns recorded meetings into tasks with owner, priority and due date, with **human-in-the-loop** approval before any write to Microsoft Planner.
- Built **multi-model routing** through OpenRouter with automatic fallback; deployed on hardened **AWS EC2** with **GitHub Actions CI/CD** and **observability** in Logfire and Langfuse.
- Designing and building a **multi-agent market-intelligence system** for a Middle East cybersecurity client: bilingual (English/Arabic) **data ingestion**, classification and schema-enforced **LLM extraction** into a **SurrealDB knowledge graph**, with human review before any CRM write.

Product Manager — *Aug - Dec 2025*

- Managed the roadmap and backlog of **Dojo**, an AI-powered platform that generates on-demand test labs, turning stakeholder needs into specifications and sprint priorities.

CED Tunisia — *Tunis, Tunisia*
Software Engineering Intern — *Feb - Aug 2026*

- Built **Collaboris**, a real-time user-presence library for enterprise web apps, shipped as two installable packages: an **Angular 19** library (npm) and an **ASP.NET Core 8** SDK (NuGet) over **SignalR** and **Redis**. Delivered in three releases under department-head review.
- Designed a five-layer identity cascade that recognises users from the host app's existing session, so nobody signs in twice.
- Built a **layered view resolver**: a scoped, debounced DOM **MutationObserver** that locates each user down to the open dialog, tab or wizard step, so co-presence is exact rather than per-route.
- Detected real user activity by wrapping the browser's **fetch** and **XMLHttpRequest** (write methods only, exclusion list, deduplication window) and multiplexed every tab onto one WebSocket with a **SharedWorker**.
- Built **PresenceAI**, the data and AI layer: **change-data capture** streaming live Redis signals into a Cosmos DB journal (**OLTP to OLAP**, Python/FastAPI), queried in natural language through a governed **Azure OpenAI** assistant over **MCP**.
- Implemented **GDPR compliance by design** (data minimisation, consent, 30-day TTL retention, fail-closed access, cost caps) and covered the AI layer with **unit tests**.
- Deployed the full stack to **Azure** (App Service, SQL Server, Azure Cache for Redis) through **Azure DevOps** build and release pipelines, migrating the API's data layer from PostgreSQL to **SQL Server**.

EDUCATION

National Diploma of Engineer, Computer Science · ESPRIT, Tunis — *Sep 2023 - Sep 2026*

Bachelor's Degree, Business Computing (E-Business), with Business Intelligence coursework · ESSECT, Tunis — *Sep 2020 - Jun 2023*

SKILLS

Backend: C#, ASP.NET Core, SignalR, Python, FastAPI, Node.js, Express, Django, REST, GraphQL, WebSockets, JWT, API gateway

Frontend: TypeScript, JavaScript, Angular, React, Next.js, Tailwind CSS, browser APIs (MutationObserver, SharedWorker), UI/UX, Figma

Data: SQL Server, PostgreSQL, MongoDB, MySQL, Redis, Cosmos DB, SurrealDB; ETL, change-data capture

Cloud & DevOps: Azure (App Service, DevOps pipelines), AWS EC2, Docker, GitHub Actions, CI/CD, Nginx, Linux, npm and NuGet packaging

Testing & AI: pytest, Storybook, ESLint, unit testing, QA; LLM integration, RAG, MCP, AI gateway

CERTIFICATIONS

Cisco CCNA (2024) · AWS Certified Cloud Practitioner (2025) · NVIDIA DLI: Building RAG Agents with LLMs (2025)

LANGUAGES

Arabic (native) · English (C1) · French (B2)